

CO3-AT3-Debugging and Optimization

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QUESTIONS

Q1

ATM Withdrawal – Exception Handling Debugging

10 marks

**Q2**Hospital Registration –
NullPointerException
Debugging

10 marks

**Q3**E-Commerce Product Search –
Array Exception Debugging

10 marks

**Q4**Bank Account – Race Condition
Debugging

10 marks

**Q5**Railway Reservation –
Synchronization Debugging

Q1 ATM Withdrawal – Exception Handling Debugging

10 marks

An ATM application is used to process cash withdrawals. The following Java program contains programming errors.

Task:

1. Run the given program and identify the errors.
2. Debug and modify the program.
3. Submit the COMPLETE corrected Java program.
4. The corrected program must handle insufficient balance, zero withdrawal, invalid input, and unexpected errors.
5. Use appropriate try-catch-finally exception handling.
6. The corrected program must execute successfully for all valid test cases.

Faulty Program:

```
import java.util.Scanner;

class ATM {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
```

10 marks

5/5 answered

```
System.out.print("Enter account balance: ");
int balance = sc.nextInt();

System.out.print("Enter withdrawal amount: ");
int amount = sc.nextInt();

try {
    int remaining = balance / amount;

    if (amount > balance) {
        System.out.println("Insufficient balance");
    } else {
        System.out.println("Withdrawal successful");
        System.out.println("Remaining balance = "
            + (balance - amount));
    }
}

catch (ArithmeticException e) {
    System.out.println("Transaction failed");
}

System.out.println("Transaction completed");
}
```

Expected correction:

The program must correctly validate the withdrawal amount before performing the transaction, handle division-by-zero and invalid input, and always display "Transaction completed" after exception handling.

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 import java.util.Scanner;
2
3 class ATM {
4     public static void main(String[] args){
5         Scanner sc = new Scanner(System.in);
6
7         System.out.println("Enter account balance :");
8         int balance = sc.nextInt();
9
10        System.out.println("Enter withdrawal amount");
11        int amount = sc.nextInt();
12        try{
13            if(amount > balance){
14                System.out.println("Insufficient balance");
15            }
16        }else{
17            System.out.println("Withdrawal successful");
18            System.out.println("Remaining balance = ");
19        }
20    }
21    catch(ArithmeticException e){
22        System.out.println("transaction failed");
23    }
24    System.out.println("Transaction completed");
25 }
```

Test Input

5000
2000

Output

Enter account balance
:
Enter withdrawal
amount:
Withdrawal successful
Remaining balance =

🕒 Pending manual review

26 }



✓ Submitted

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